# Sergey Mozgovoy

**C#/.NET Developer · AgTech · GIS · Data**

Moscow · office / hybrid / remote • commercial development since 2021 • agronomy and variety testing since 2008

## PROFILE

C#/.NET developer with commercial experience since 2021 and more than ten years in agriculture: variety testing, agrochemistry and crop production. I focus on the intersection of software development, GIS and agricultural data processing.

I own the full cycle: requirements analysis, data model and application design, implementation, rollout and support. I build ASP.NET Core backend services, WinForms desktop applications, integration layers to legacy systems and reporting automation tools.

## EXPERIENCE

### Backend Developer

07.2024 – 02.2026

Altura

- Backend services of an information and analytics platform on C#/.NET and ASP.NET Core; designed REST APIs, data models and service structure independently.
- Designed and implemented the backend of the geo-globus mapping service: application, API and a PostgreSQL/PostGIS database.
- EF Core, LINQ, async/await, DI; PostgreSQL migrations, transactions and indexes, N+1 elimination and query optimisation.
- Inter-service messaging via RabbitMQ/EasyNetQ, background jobs with Quartz.NET, Polly policies (retry, circuit breaker, timeout).
- Integrations with external HTTP APIs, OpenSearch (mapping, batch writes, search, aggregations), Docker and Kubernetes manifests, GitLab CI/CD.

`C#` `ASP.NET Core` `EF Core` `PostgreSQL/PostGIS` `RabbitMQ` `OpenSearch` `Quartz.NET` `Docker`

### Chief Programmer

09.2021 – 07.2024 · 0.5 FTE since 08.2024

FSBI «Gossortkomissiya» (state variety-testing commission)

- Full cycle of development and support for state variety-testing systems: head office, 45 branches, trial sites and laboratories.
- Built an integration layer between the legacy Centura system and .NET applications, plus a generator of object models and CRUD code — over 600,000 generated lines.
- Developed and maintain the «SPRAVKA» system: aggregation and analysis of variety-testing and laboratory results, including historical data of about 30 years.
- Designed a PostgreSQL database with more than 100 related tables and utilities to migrate data from Centura.
- Trial-site planning, annexes to orders and registers for the FSIS «Seed Production» system, two-way data exchange.
- Automated analytical queries, variety descriptions, Excel and Word documents and expert-commission materials; some operations became hundreds of times faster.

`C#` `.NET Framework` `WinForms` `PostgreSQL` `EF Core` `ADO.NET` `XML`

### Lead Agronomist

12.2008 – 03.2017 · 06–10.2017 · 08.2019 – 10.2022

Krasnoturansky state variety-testing site

- Full cycle of field variety trials: planning and laying out plots, sowing, phenological observations, yield accounting.
- Statistical processing and randomisation; built applied software for experimental data processing.

### Lead Agrochemist → Head of Soil Fertility Monitoring

03–06.2017 · 10.2017 – 08.2019

Minusinsk agrochemical station

- Organised agrochemical and eco-toxicological surveys of more than 200,000 hectares.
- Fertility monitoring, soil and leaf diagnostics, agrochemical cartograms, fertiliser recommendations.
- Built the «GSAS Minusinsk» system with data migration from FoxPro, plus «Nitrates» and nutrient-balance applications.

### Agronomist

01.2021 – 09.2021

ZAO «Iskra Lenina»

- Crop production on about 6,000 hectares: production cycle, process maps, crop rotations, fertilisers, plant protection, field mapping.

## PROJECTS

- **Agrometeorological information system** — an archive of about 20 years of observations: temperature sums, precipitation, snow cover, hydrothermal coefficient, frost risk, PAR. C#/.NET, PostgreSQL/PostGIS, RabbitMQ, Quartz.NET, Playwright.
- **Brainstable.Applications** — official bulletin assembly: 13 CSV exports, 15 tables, 16 Word templates merged into a single `.docx` with title page and pagination.
- **BarleyGWAS** — phenotypic and genotypic data processing for the VIR barley collection: quality control, statistics, GWAS visualisation. C#/.NET, WinForms, R.
- **GeoPlots and GeoHistory** — interactive Leaflet maps: trial sites and abandoned villages with historical data.

## SKILLS

C# · .NET 10 · ASP.NET Core · WinForms · REST API · EF Core · LINQ · async/await · SOLID

PostgreSQL · PostGIS · SQL · RabbitMQ · OpenSearch · NetTopologySuite · GeoJSON · ArcGIS · QGIS

variety testing · field trials · agrochemistry · agrometeorology · statistics · R

Git · GitLab CI/CD · Docker · Kubernetes · Swagger · Polly · Quartz.NET

## EDUCATION

### PhD programme, Plant Breeding and Genetics

VIR — N. I. Vavilov Institute of Plant Genetic Resources

2024 — present · GWAS of agronomically valuable barley traits

### MSc, Agronomy

Krasnoyarsk State Agrarian University

2018

### Physics and Computer Science

Khakass State University

incomplete higher education

## LANGUAGES

| | |
|---|---|
| Russian | native |
| English | technical reading, professional literature |

## CONTACT

github.com/brainstable

brainstable.github.io/en

Issues — questions and ideas

Email and phone on request.